

The equation can be solved using cyclotomy and quadratic residues. A partial solution was found by Dirichlet using this method, building upon the work of Gauss [3]. In this section, we build upon Dirichlet's work, explicitly writing the solution and using the modern machinery of Galois Theory to streamline the approach. Again, we let  $p$  be an odd prime, and define  $p^* = (-1)^{\frac{p-1}{2}}p$ ,  $i = \sqrt{-1}$ , and start by introducing an important lemma.

**Lemma 3.**  $\left\{ \begin{array}{l} q_1(x) = 2 \prod_{\substack{1 \leq k < p \\ \left(\frac{k}{p}\right)=1}} (x - \zeta^k) = f(x) + \sqrt{p^*}g(x) \\ q_{-1}(x) = 2 \prod_{\substack{1 \leq k < p \\ \left(\frac{k}{p}\right)=-1}} (x - \zeta^k) = f(x) - \sqrt{p^*}g(x) \end{array} \right. \quad \text{where } f(x), g(x) \text{ are polynomials in } \mathbb{Z}[x].$

*Proof.* Note that the product of the 2 above polynomials (on the left-hand side) is  $4 \prod_{1 \leq k < p} (x - \zeta^k) = 4m_p(x) \in \mathbb{Z}[x]$ . It is therefore fixed by any Galois automorphism in  $\text{Gal}(K : \mathbb{Q})$ . Now taking  $\theta = \zeta^{\frac{p^2-1}{8}} \prod_{k=1}^{\frac{p-1}{2}} (1 - \zeta^k)^2$ , we see that  $\theta^2 = p^*$  since  $(-1)^{\frac{p^2-1}{8}} \equiv \left(\frac{2}{p}\right) \pmod{2}$ , and trivially  $\theta \in \mathcal{O}_K$ . So  $\sqrt{p^*} \in \mathcal{O}_K$ . Now an automorphism  $\sigma$  in the Galois group fixes  $p^*$  if and only if  $\sigma$  is a square. But this is if and only if  $\sigma$  fixes all (and only) the  $\zeta^k$  such that  $k$  is a quadratic residue modulo  $p$ . So  $\prod_{\substack{1 \leq k < p \\ \left(\frac{k}{p}\right)=1}} (x - \zeta^k) \in L[x]$  where  $L = \mathbb{Q}(\sqrt{p^*})$ . All the coefficients in  $L[x]$  are of the form  $a + b\sqrt{p^*}$  where  $a$  and  $b$  are both rational, and  $\frac{1}{2}$  an algebraic integer (allowing for the fact that  $p^* \equiv 1 \pmod{4}$ ). The coefficients of  $2 \prod_{\substack{1 \leq k < p \\ \left(\frac{k}{p}\right)=1}} (x - \zeta^k)$  are therefore rational algebraic integers and thus in  $\mathbb{Z}$ . We can now expand  $q_1(x)$  and rewrite it as  $q_1(x) = f(x) + \sqrt{p^*}g(x)$  where  $f(x), g(x)$  are polynomials in  $\mathbb{Z}[x]$ .

A similar argument shows that  $q_{-1}(x) \in L[x]$ . Now let  $\tau$  be the Galois automorphism in  $\text{Gal}(K : \mathbb{Q})$  defined by  $\tau(\sqrt{p^*}) = -\sqrt{p^*}$  (noting that  $K : L : \mathbb{Q}$  is a tower of fields). Then by the above, and since  $\tau^2$  must fix  $q_1(x)$ , we must have that  $\tau(\zeta^k) = \zeta^l$  where  $\left(\frac{k}{p}\right)\left(\frac{l}{p}\right) = -1$ . So since  $\tau$  is a Galois automorphism over  $K$ , we must have  $\tau(q_1(x)) = q_{-1}(x)$ . This yields that  $q_{-1}(x) = f(x) - \sqrt{p^*}g(x)$ .  $\square$

We will primarily consider the case where  $d$  is an odd prime. Pell's Equation then becomes

$$(3.1) \quad x^2 - py^2 = 1$$